

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

- LO 4.3.A** — Identify the direction of planar motion at any value of the parameter by determining whether $x(t)$ and $y(t)$ are increasing or decreasing at that moment. _____
- LO 4.3.A** — Explain why the same curve can be traversed in different directions by different parametrizations. _____
- LO 4.3.A** — Compute the average rate of change of $x(t)$ and $y(t)$ over $[t_1, t_2]$ and use their ratio to find the slope of the parametric curve between two points. _____

Vocabulary & Key Concepts

direction of motion: Determined by whether each component is _____ or _____. If $x(t)$ is increasing $\Rightarrow$ moving _____; if decreasing $\Rightarrow$ moving _____. If $y(t)$ is increasing $\Rightarrow$ moving _____; if decreasing $\Rightarrow$ moving _____.

ARC of $x(t)$: The average rate of change of $x(t)$ over $[t_1, t_2]$: $\frac{x(t_2) - x(t_1)}{t_2 - t_1}$

ARC of $y(t)$: The average rate of change of $y(t)$ over $[t_1, t_2]$: $\frac{y(t_2) - y(t_1)}{t_2 - t_1}$

slope of parametric curve: The slope of the _____ line connecting two points on the curve: $\frac{\Delta y}{\Delta x} = \frac{\text{ARC of } y(t)}{\text{ARC of } x(t)}$ provided ARC of $x(t) \neq 0$.

reparametrization: A different pair of component functions that traces the _____ set of points in the plane, possibly in a _____ direction.

Hook

Entry Question: Two drones follow the same figure-eight path — Drone A clockwise, Drone B counterclockwise.

Q1. Can two different parametric functions trace the same path? What would have to be different about them? _____

Q2. How do you determine which direction a particle is moving horizontally at a specific value of t ? _____

LO 4.3.A — Direction of Planar Motion (EK 4.3.A.1–A.2)

Key rule: Analyze $x(t)$ and $y(t)$ *independently*.

If $x(t)$ is increasing	particle moves _____
If $x(t)$ is decreasing	particle moves _____
If $y(t)$ is increasing	particle moves _____
If $y(t)$ is decreasing	particle moves _____

Example: Let $f(t) = (t^2 - 2t, 3t - t^2)$ for $0 \leq t \leq 4$.

Step 1 — Build a table of values.

t	$x(t) = t^2 - 2t$	$y(t) = 3t - t^2$	(x, y)
0	_____	_____	_____
1	_____	_____	_____
2	_____	_____	_____
3	_____	_____	_____
4	_____	_____	_____

Step 2 — Direction at $t = 0.5$.

Compare $x(0)$ and $x(1)$: $x(0) = \underline{\hspace{2cm}}$, $x(1) = \underline{\hspace{2cm}}$.

Since $x(1) \underline{\hspace{2cm}} x(0)$, the function $x(t)$ is _____ near $t = 0.5$.

$\Rightarrow$ particle is moving _____

Compare $y(0)$ and $y(1)$: $y(0) = \underline{\hspace{2cm}}$, $y(1) = \underline{\hspace{2cm}}$.

Since $y(1) \underline{\hspace{2cm}} y(0)$, the function $y(t)$ is _____ near $t = 0.5$.

$\Rightarrow$ particle is moving _____

Step 3 — Direction at $t = 2.5$.

Compare $x(2)$ and $x(3)$: is x increasing or decreasing? _____

$\Rightarrow$ particle is moving _____

Compare $y(2)$ and $y(3)$: is y increasing or decreasing? _____

$\Rightarrow$ particle is moving _____

LO 4.3.A — Same Curve, Different Parametrizations (EK 4.3.A.3)

Key idea: The same set of points in the plane can be traced by different parametric functions — possibly in different directions.

Example: Let $f(t) = (t, t^2)$ and $g(t) = (3-t, (3-t)^2)$ for $0 \leq t \leq 3$.

	Robot A: $f(t) = (t, t^2)$		Robot B: $g(t) = (3-t, (3-t)^2)$	
t	(x, y) for f	direction	(x, y) for g	direction
0	_____		_____	
1	_____		_____	
2	_____		_____	
3	_____		_____	

Do f and g trace the same set of points? _____

Are they moving in the same direction? _____

LO 4.3.A — Slope of the Curve via Average Rates of Change (EK 4.3.A.4)

Formula: Over the interval $[t_1, t_2]$:

$$\text{slope of curve} = \frac{\Delta y}{\Delta x} = \frac{y(t_2) - y(t_1)}{x(t_2) - x(t_1)} \quad (\text{provided } x(t_2) \neq x(t_1))$$

Example (continued): Use $f(t) = (t^2 - 2t, 3t - t^2)$. Find the slope of the curve from $t = 1$ to $t = 3$.

$$\begin{array}{lll} x(1) = \underline{\hspace{2cm}} & x(3) = \underline{\hspace{2cm}} & \Delta x = \underline{\hspace{2cm}} \\ y(1) = \underline{\hspace{2cm}} & y(3) = \underline{\hspace{2cm}} & \Delta y = \underline{\hspace{2cm}} \end{array}$$

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{\underline{\hspace{2cm}}}{\underline{\hspace{2cm}}} = \underline{\hspace{2cm}}$$

Interpretation: The secant line connecting the points on the curve at $t = 1$ and $t = 3$ has slope _____.